

Dobbertin (1999), part II: the gadget was a category

How the three faces **L** **F** **C** become one bridge, one invariant, and five readings — machine-checked

Categorical continuation of “one gadget, many theorems” — Lean / Mathlib `CategoryTheory`

Where part I left off. The picture book “one gadget, many theorems” argued that every result of Dobbertin’s paper is the *same small gadget* with three faces: **L** the **linearized trace** $L_k(x) = \sum_{i < k} x^{2^i}$; **F** **Frobenius** $x \mapsto x^2$ (doubling on exponents); **C** **cyclotomic-coset** book-keeping ($\times 2 \bmod 2^n - 1$). **This part answers the follow-up question:** is that “one gadget” story consistent with what actually got *formalised* as structural / categorical patterns? **Yes** — and more: the three faces are not just a slogan, they are literally *one structure, one invariant, and one equivalence relation* in Lean. Every green ✓ box below is sorry-free in RequestProject/Kasami/.

0. The one-line verdict

The part-I gadget and the categorical formalisation are **the same object**. The gadget’s data **L** is exactly the field `Pattern.star` (the identity $(\star)$); its symmetry **F** is exactly a `ConjHom` / self-`SiteEquiv` (`frobeniusConjHom`); and its bookkeeping **C** is exactly the *bridge-stable* `Pattern.width` read across sites. “One gadget seen from many angles” = “one `Pattern`, many `SiteEquiv`-related readings”.

Part-I face	Categorical structure (Lean)	Status
L linearized trace L_k , identity $(\star)$	<code>Pattern</code> / <code>ConjSquare</code> , field <code>star/comm</code>	✓
F Frobenius $x \mapsto x^2$	<code>frobeniusConjHom</code> (auto of the square); <code>SiteEquiv</code>	✓
C cyclotomic / coset counting	<code>Pattern.width</code> , <code>width_transport</code> , difference-set reading	✓

1. The gadget is a structure: the pattern $(\star)$

Dobbertin’s Theorem 4 identity — the Kasami collision map conjugated to the Müller–Cohen–Matthews companion —

$$(\star) \quad \underbrace{(s_k(t) + 1)}_{\text{diff}} \cdot \underbrace{(L_1 t)^{2^k}}_{\text{denom}} = \underbrace{(L_k t)^{2^k + 1}}_{\text{numer}}$$

is packaged *as data* on any commutative ring:

```
structure Pattern (R)
  diff : R → R
  denom, numer : R → R
  star : ∀ t, diff t * denom t = numer t
```

instance

```
kasamiPattern k ✓
  diff = s_k + 1, denom = (L_1)^{2^k},
  numer = (L_k)^{2^k + 1}, star = mcm_square
```

Lean: `SiteBridge.Pattern`, `SiteBridge.kasamiPattern`, `ConjugacyPattern.mcm_square` (the identity $(\star)$, `char.2`). ✓

2. The bridge is transport: `SiteEquiv` is an equivalence relation

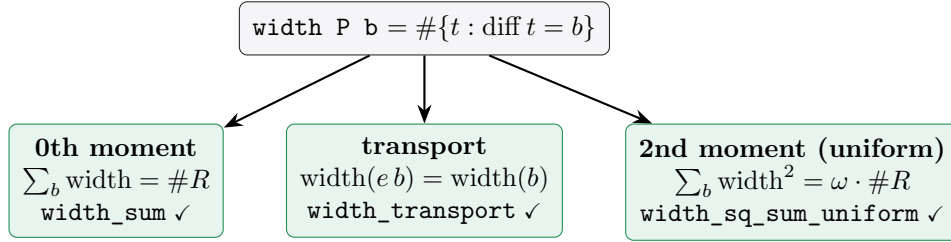
The “Caramello / Morita bridge” move is: carry a pattern along a ring isomorphism $e : R \simeq S$. The very same $(\star)$ then holds in the new carrier *for free* (`transport_star`). Two patterns are a *site equivalence* when some e carries one to the other; this is proved reflexive, symmetric and transitive — the formal skeleton of “these sites present the same classifying topos”.

$$\begin{array}{ccc}
(R, P) & \xrightarrow[\text{transport}]{e : R \simeq S} & (S, P.\text{transport } e) \\
(\star) \downarrow & & \downarrow (\star) \text{ for free} \\
\text{diff} \cdot \text{denom} = \text{numer} & \xleftarrow{\text{same law}} & \text{diff}' \cdot \text{denom}' = \text{numer}'
\end{array}$$

Lean: `Pattern.transport, transport_star, transport_refl, transport_trans; SiteEquiv with SiteEquiv.refl / .symm / .trans. ✓`

3. One invariant for all sites: `width` is bridge-stable

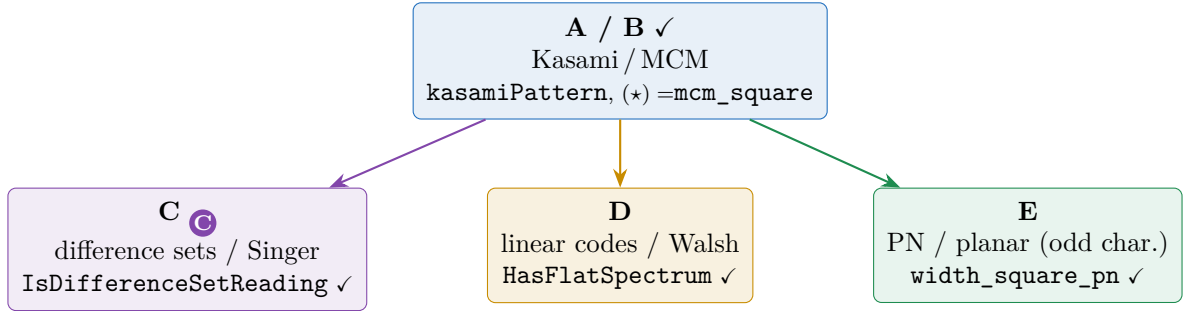
Every downstream reading of the paper — APN, difference set, Walsh spectrum — reads exactly one number off the pattern: the *width* $b \mapsto \#\{t : \text{diff } t = b\}$. The key structural theorem is that *width* is *literally the same* through every site equivalence, so an invariant may be computed in whichever site is easiest.



This is the abstract shell of the project’s `EWIGO.wid_partition` and `EWIGO.coll_of_uniform`: the same two moments, now bridge-stable. ✓

4. Five sites, one width: A/B/C/D/E as readings

The paper’s chapters become *readings* of the one width invariant; each is obtained “for free” because *width* is bridge-stable.

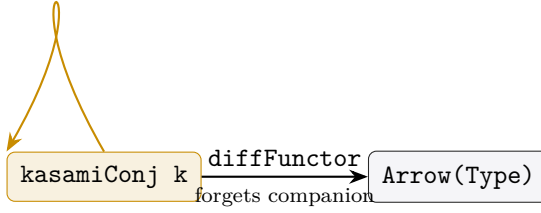


Each reading’s defining predicate is *proved bridge-stable* (`isDifferenceSetReading_transport, hasFlatSpectrum_transport, squarePattern_siteEquiv`); site E’s optimum width $\equiv 1$ is outright *proved* (`width_square_pn, pnIdentification_holds`). ✓

5. The bridge is a category (the face)

Over a fixed carrier the patterns form a genuine `CategoryTheory.Category`: objects are conjugacy squares, morphisms are intertwining ring endomorphisms (the *2-morphisms* of the connection). A forgetful functor to the arrow category keeps only `diff` and is *non-faithful* — the companion is real extra data. Frobenius $x \mapsto x^2$ is an *automorphism* of the Kasami square, and its powers are the finitely many Galois-equivalent contexts.

frobeniusConjHom (F)



instance : Category (ConjSquare F) ✓
 ConjHom, .id, .comp ✓
 diffFunctor, diff_not_injective ✓
 frobeniusConjHom (F auto) ✓

6. The whole paper, sitting on the structural machinery

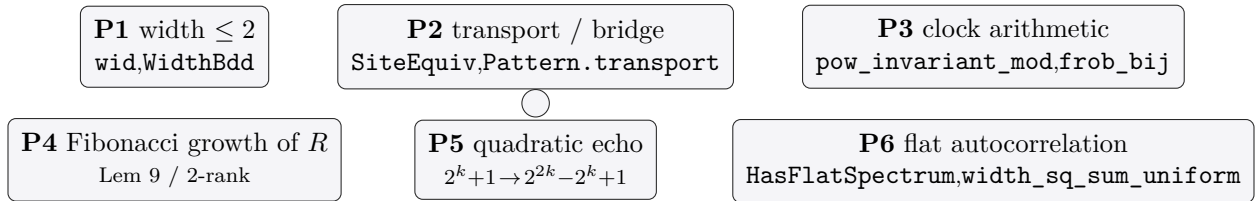
Each numbered result of Dobbertin (1999) read through the structural patterns. ✓ = machine-checked structural statement; ○ = stated as an honest open Prop (never an axiom).

Result (Dobbertin 1999)	Faces	Structural home (Lean) — status
Thm 4 — MCM ⇔ Kasami bridge	L	kasamiPattern.star = mcm_square ✓
“ — same law in every context	L F	kasamiPattern.siteEquiv, transport_star ✓
Thm 1/3 — permutation criterion	L F	frobeniusConjHom intertwiner ✓
Cor 2 — Kasami APN = width ≤ 2	L F C	HasFlatSpectrum _ 2 ○ CodeWalshIdentification
Thm 5/6, Cor 7 — explicit inverse R	F	Frobenius iteration; Theorem6.* ✓
Thm 8 — $B = \{\text{Tr } R = 0\}$	F C	Pattern.fibre / trace reading
Cor 10/11/12 — 2-rank, $k'=3$	C	pow_invariant_mod (coset count) ✓
§4 Conjecture — B^* diff. set	C	IsDifferenceSetReading ○ DifferenceSetIdentification
Rigidity (only Kasami works)	L F C	RigidityConjecture ○
Count of contexts = n	F	ContextCountConjecture ○
PN analogue (odd char.)	C	width_square_pn ✓

Faithfulness. The open rows are recorded exactly as the paper leaves them — as statements, not assumptions. Following the project’s discipline they are Lean Props (DifferenceSetIdentification, CodeWalshIdentification, RigidityConjecture, ContextCountConjecture), so they can later be *proved or refuted*; **none is asserted as an axiom**. The one that is not open — the PN reading — is proved (pnIdentification_holds).

7. The six recurring patterns, as one diagram

Part I of the companion note (dobbertin-story) named six recurring structural patterns P1–P6. They are exactly the ways the single Pattern propagates:



P1 = the optimum of width; P2 = transport/SiteEquiv; P3 = Frobenius (F) (frob_bij, pow_invariant_mod); P4/P5 = the exponent recursions carried by (F); P6 = the flat-spectrum (HasFlatSpectrum) reading whose second moment is width_sq_sum_uniform. So P1–P6 are L F C again — consistent with part I.

8. Verdict

Consistent, and sharper. Part I’s “one gadget, three faces” is not merely a picture: in the Lean development the gadget is one `structure` (`Pattern`), its symmetries form one `Category` with a `SiteEquiv` equivalence relation, and its content is one bridge-stable invariant (`width`) read through five sites. The paper’s proved results sit on that machinery as $\checkmark$ -theorems; its open results sit on it as honest `Props`. The categorical layer is therefore a faithful, structural re-telling of Dobbertin (1999) — the same story, built wherever possible along the shortest structural path.